Fast Flow Simulations

Multigrid solvers and reduced precision for 2D laminar channel flow in OpenFOAM

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What I'm doing, and why

Simulating a flow means chopping the space into millions of tiny cells and solving equations in each one. Most of the runtime goes into one step that repeats every timestep: solving a big system of equations for pressure. This is expensive.

Two ways to make that step cheaper:

- **Better solver:** multigrid (GAMG) instead of PCG
- **Less precision:** 32-bit numbers instead of 64-bit, so half the data to move

The question: How much computational cost can be removed without compromising accuracy or solver convergence?

Procedure:

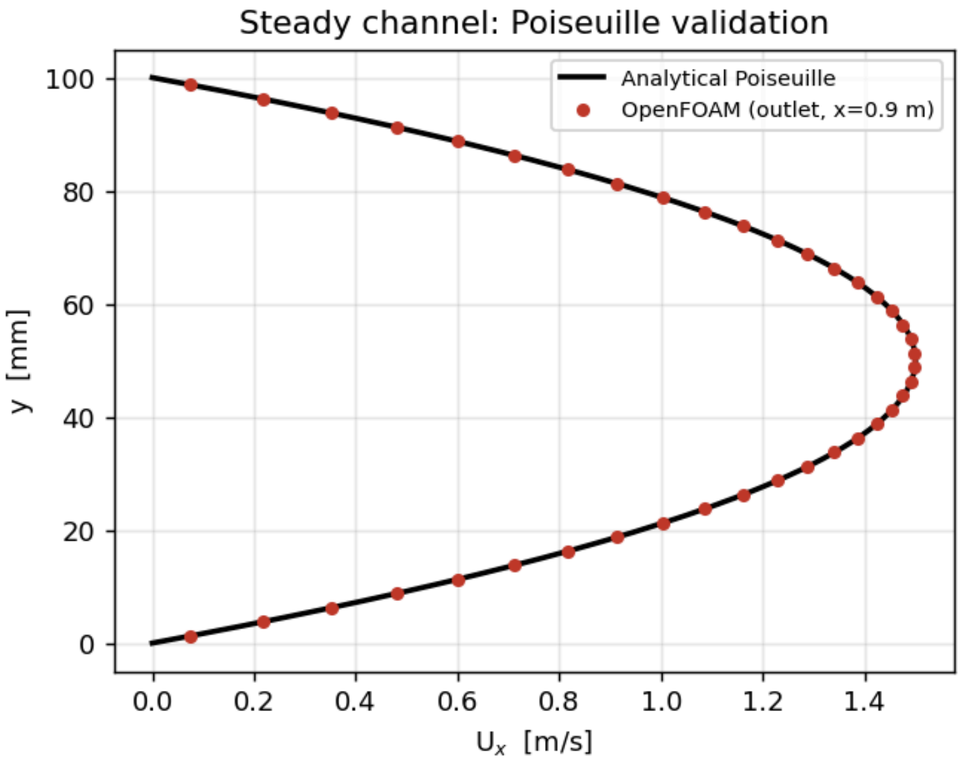
1. **Establish and validate a benchmark**
2. **Compare Solvers: PCG & GAMG. Show which solver is FASTER** **(RESULT 1)**
3. **Establish mesh independence: Find the COARSEST mesh we can work on** **(RESULT 2)**
4. **Compare Precision: Double precision (fp64) → Single Precision (fp32)** **(RESULT 3)**
5. **Apply our setup to more complex inlet conditions** **(RESULT 4)**
6. **Discussion: Any interesting results**
7. **Summary and next steps**

Validating benchmark and controlled timing protocol

Plane channel flow has an exact analytical solution, the **Poiseuille parabola**, so the simulation can be **compared against a known-correct answer**, rather than only against other runs of the same code.

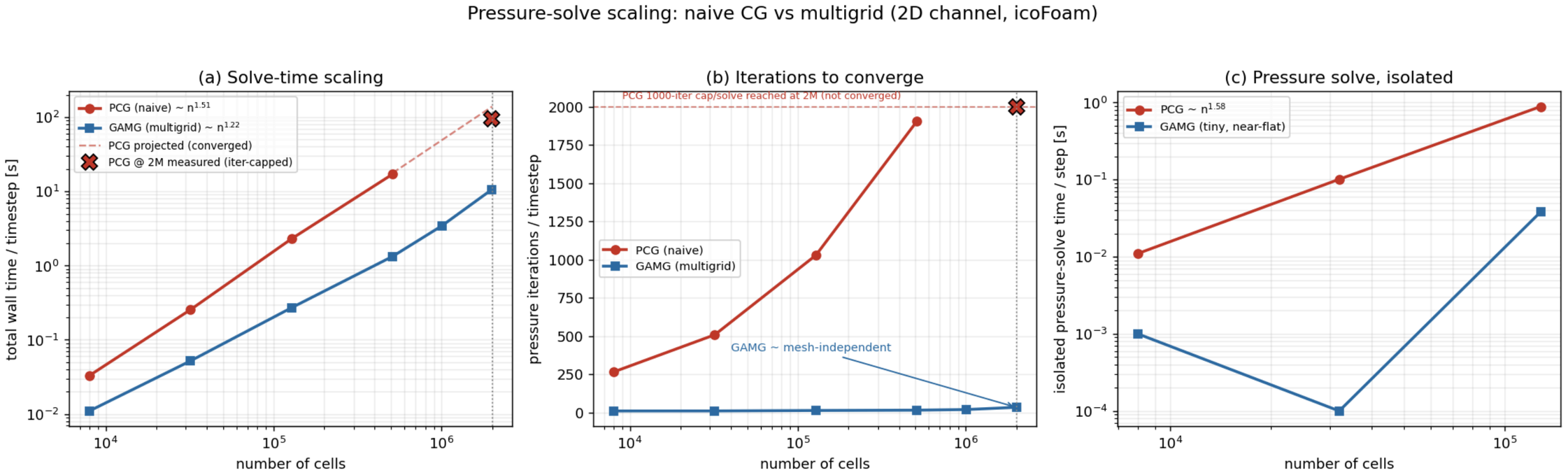
Conditions:

- 2D channel, Re_H=100, laminar
- Analytical Poiseuille peak: 1.5 ms^{-1}
- Simulated profile agrees within 0.08%
- Three alternating timing repeats; median and spread reported



Simulated points sit on the exact parabola.

PCG vs GAMG



Cost growth

- Plain solver (PCG): n^1.51
- Multigrid (GAMG): n^1.22

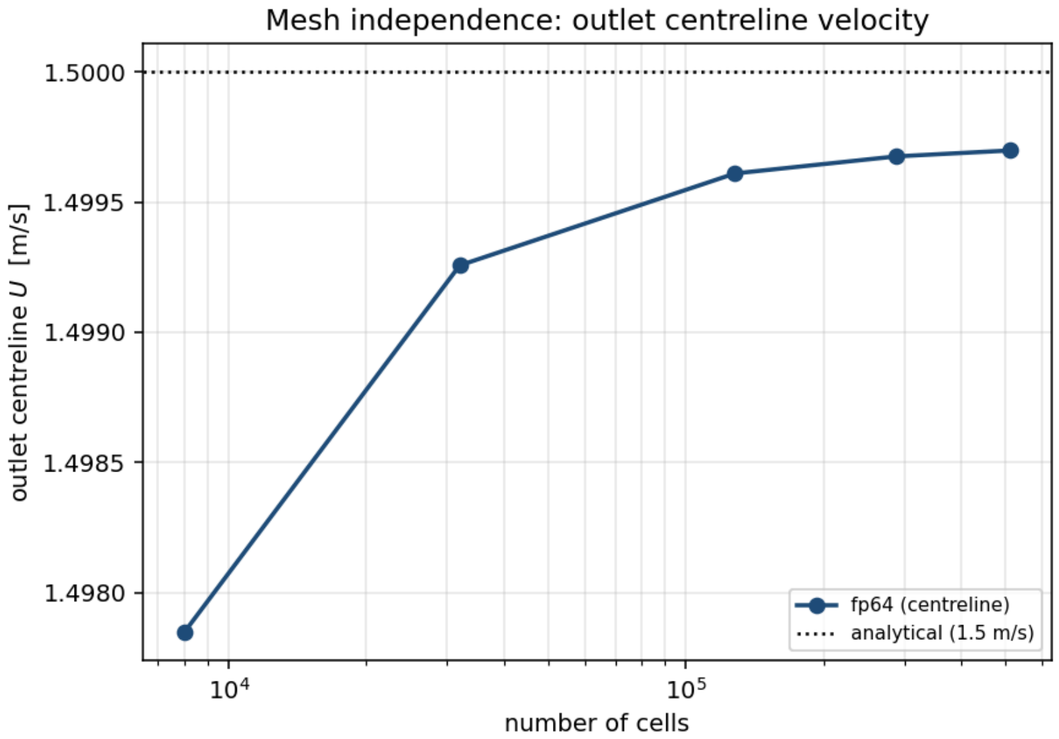
Iterations per step (8k → 2M cells)

- PCG: 265 → 1,905
- GAMG: 10 → 33

At 2 million cells

- GAMG: ~11 s per step
- PCG: ~20 min per step - extrapolated

How fine does the mesh need to be?



The analytical error plateaus near 0.018% because the outlet profile is approximately 99.98%, rather than fully, developed. because of Physics.

Successive changes:

0.094% → 0.024% → 0.004% → 0.0015%

Selected mesh

Approximately 128k cells satisfies the 0.01% criterion

Verification

Observed spatial order ≈ 2.0

A mesh of approximately 128k cells satisfies the 0.01% criterion.

RESULT 3 — PRECISION

fp32 (single precision) vs fp64 (double precision)

- Accuracy: Where both converge, fp32 and fp64 agree to 3×10^{-7} ms^{-1}.
- Memory: fp32 saves 14–28%, on every comparison
- **Speed:**

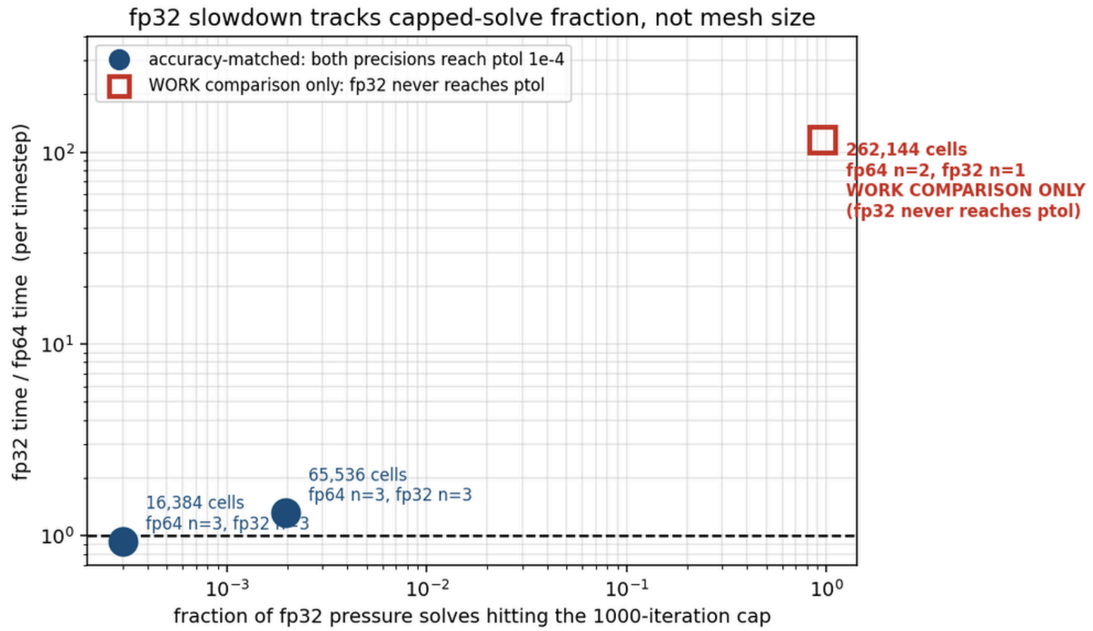
Mesh	Pressure solves reaching the 1,000-iteration cap	fp32 vs fp64 speed
16,384 cells	never (0.00%)	**6.7% FASTER**
65,536 cells	2 solves out of 1,024 (0.20%)	**31% slower**
262,144 cells	almost always (95.31%)	**fp32 failed to reach tolerance***

The immediate predictor of slowdown is the fraction of pressure solves reaching the iteration cap, rather than mesh size alone.

As the fp32 convergence floor rises, the requested pressure tolerance eventually becomes unreachable.

* *At 262k cells fp32 took approximately 116× more pressure-solver time*

Investigating fp32 slowdown



Slowdown tracks how often the solver gives up, not mesh size.

Every capped pressure solve costs 1,000 iterations.

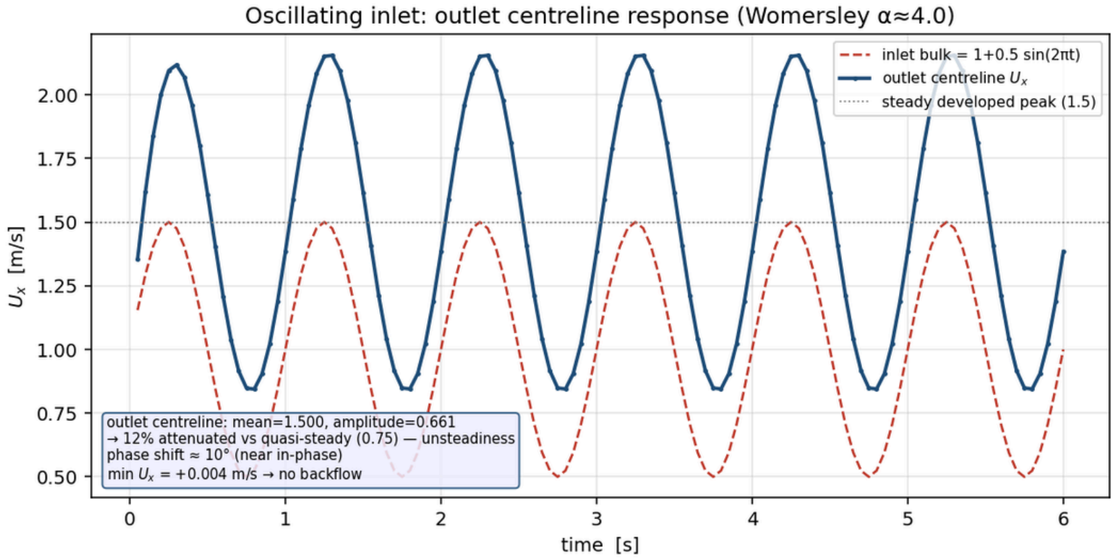
- At 65k cells, only 2 of 1,024 pressure solves reached the cap.
- Those two solves contributed 55% of all fp32 pressure iterations.
- Excluding them, fp32 and fp64 perform similar total work.
- fp32 remains cheaper per iteration: 2.99 vs 3.46 ms.

PCG cross-check: no capped solves, but fp32 required 9% more iterations and was 7.7% slower.

The slowdown is caused by capped solves, not directly by mesh size

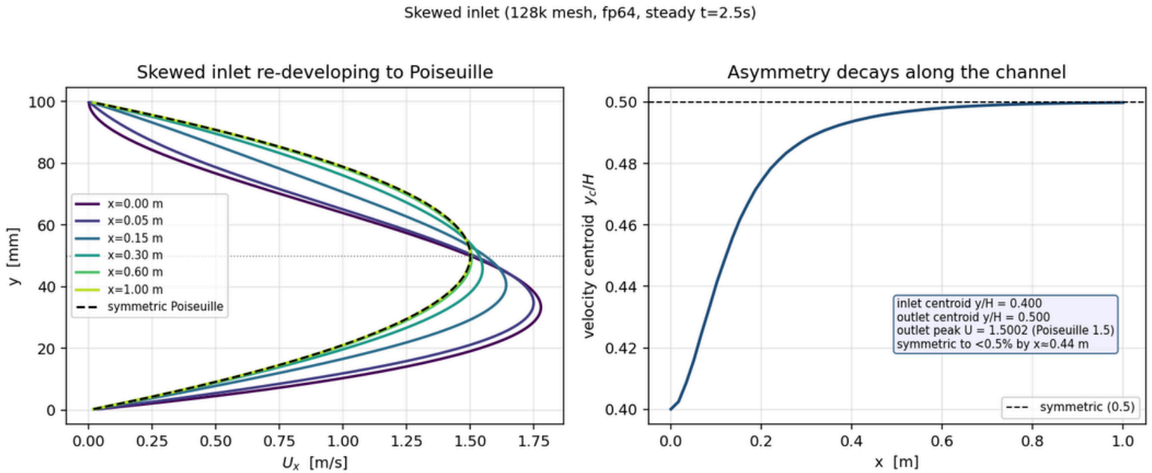
Applying the optimised setup to more complex inlet conditions

Oscillating inlet (α ≈ 4)



The outlet oscillation is attenuated because the profile cannot fully redevelop during each cycle. No backflow occurred.

Skewed inlet



The velocity centroid moves from 0.40 to 0.50 of the channel height, showing recovery towards the symmetric Poiseuille profile.

DISCUSSION

Two machines, opposite answers

on Arm	on Apple Silicon (65k cells)
fp32 21.4% FASTER	**fp32 31% SLOWER**
big memory saving	big memory saving

Both results are explained by the same convergence mechanism.

When the solver reaches its target, fp32's lower per-iteration cost produces a speed benefit. When it cannot reach the target, capped solves outweigh that benefit. My smaller-mesh Apple Silicon result was also faster in fp32, consistent with the Arm measurement.

Memory reduction was observed on both platforms.

Speed depends on the hardware (for now).

Results summary

1. **Multigrid:**

GAMG changes pressure-solver scaling from $n^{1.51}$ to $n^{1.22}$ by keeping iterations nearly mesh-independent.

2. **Accuracy and memory:**

Where fp32 converges, it matches fp64 to 3×10^{-7} ms^{-1} and reduces peak memory by 14–28%.

3. **Runtime:**

fp32 speed is conditional: capped pressure solves can outweigh its lower per-iteration cost.

Next hypothesis

Mixed precision: retain the difficult pressure solve in fp64 while using fp32 elsewhere, followed by testing on Arm/Isambard.